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Date:

APP 1: Login Page

Aim: Design a Login page with username and password using Linear Layout and Toast Valid and Invalid credentials.

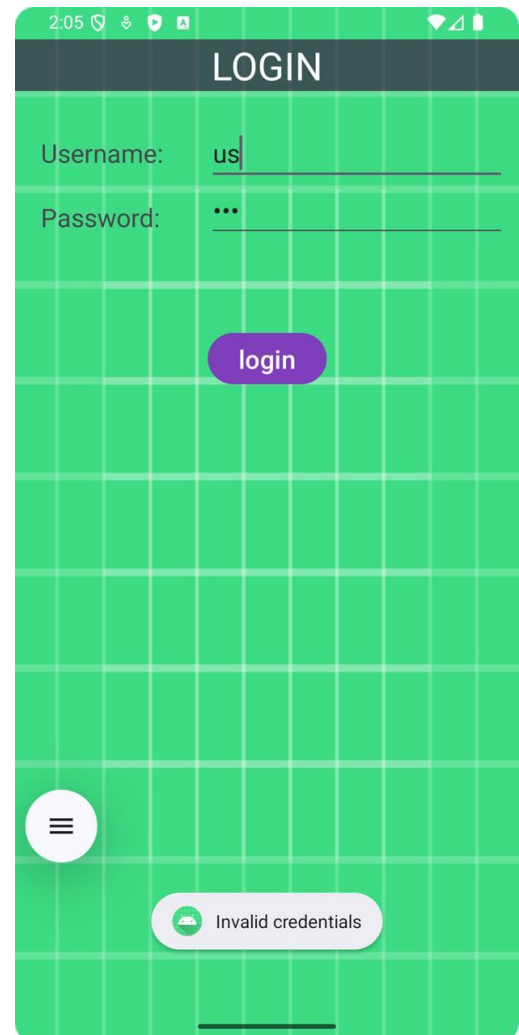
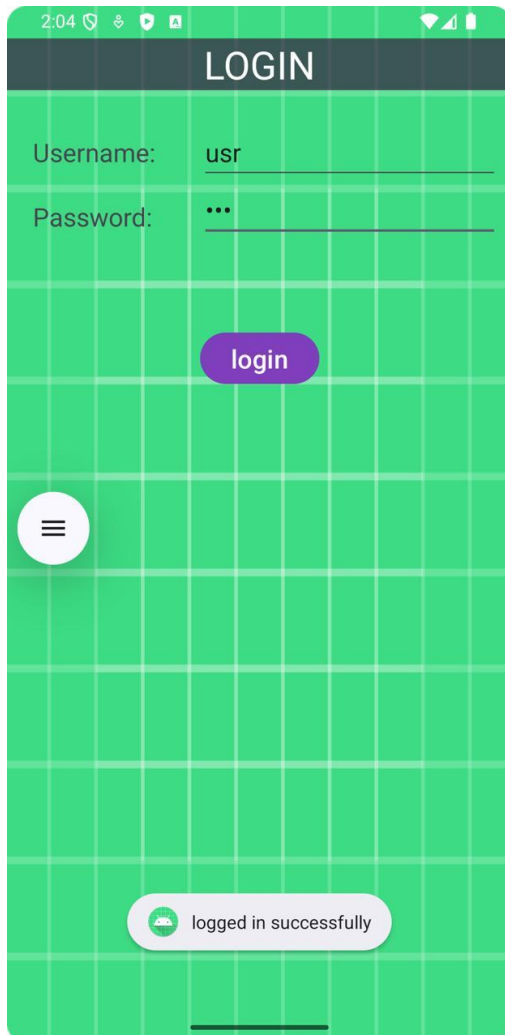
Program Code:

MainActivity.java

```
package com.example.myapplication;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    Button b;
    EditText u,p;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        b=(Button) findViewById(R.id.b);
        u=(EditText) findViewById(R.id.user);
        p=(EditText) findViewById(R.id.pass);
        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String user=u.getText().toString();
                String pass=p.getText().toString();
                if (user.equals("usr")&& pass.equals("123"))
                    Toast.makeText(getApplicationContext(),"logged in
successfully",Toast.LENGTH_SHORT).show();
                else
                    Toast.makeText(getApplicationContext(),"Invalid
credentials",Toast.LENGTH_SHORT).show();
            }
        })
    }
}
```



Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:background="@drawable/ic_launcher_background">
    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <TextView
            android:id="@+id/l"
            android:layout_width="1200dp"
            android:layout_height="wrap_content"
            android:layout_alignParentEnd="true"
            android:layout_marginTop="25dp"
            android:layout_marginEnd="0dp"
            android:background="#B23B1E40"
            android:text="LOGIN"
            android:textAlignment="center"
            android:textColor="#ffffff"
            android:textSize="30dp" />

        <TextView
            android:id="@+id/u"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Username:"
            android:layout_marginTop="100dp"
            android:layout_marginLeft="20dp"
            android:textSize="20dp"
            />

        <TextView
            android:id="@+id/textView3"
            android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:text="Password:"
        android:layout_marginTop="150dp"
        android:layout_marginLeft="20dp"
        android:textSize="20dp"
        />

        <Button
            android:id="@+id/b"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginLeft="150dp"
            android:layout_marginTop="250dp"
            android:backgroundTint="#DF803BBC"
            android:text="login"
            android:textSize="20dp" />

        <EditText
            android:id="@+id/user"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="90dp"
            android:layout_marginLeft="150dp"
            android:textSize="20dp"
            android:ems="10"
            android:inputType="text"
            android:text="" />

        <EditText
            android:id="@+id/pass"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="135dp"
            android:layout_marginLeft="150dp"
            android:textSize="20dp"
            android:ems="10"
            android:inputType="textPassword" />
    </RelativeLayout>
</LinearLayout>

```

Result: The program has executed successfully and output is verified

Date:

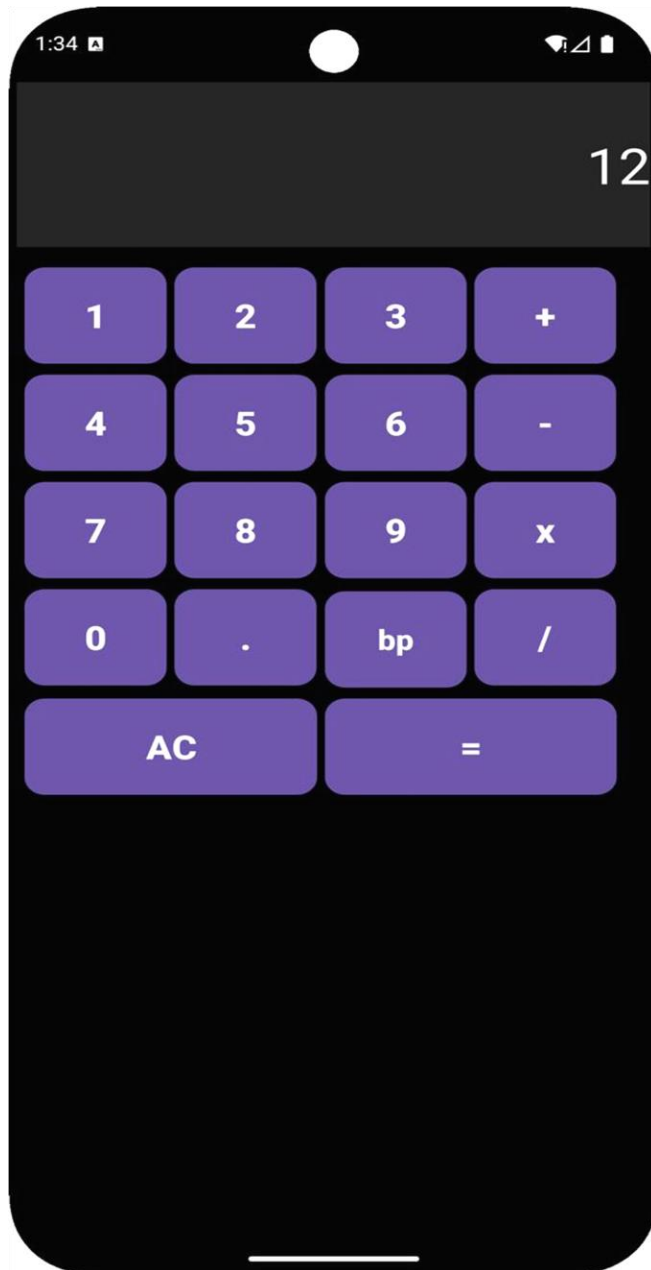
APP 2: Simple Calculator

Aim: Create a simple calculator to perform arithmetic operations

Program Code:

MainActivity.java

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity
{
    EditText t1;
    Button bac;
    String firstNumber="",operator="";
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        t1=(EditText) findViewById(R.id.t1);
        bac=(Button) findViewById(R.id.bac);
        int b[]=new int[17];
        for(int i=0;i<=16;i++)
        {
            String s="b"+(i);
            int rid=getResources().getIdentifier(s,"id",getPackageName());
            b[i]=rid;
        }
        for(int id:b)
        {
            Button button=findViewById(id);
            if(button !=null)
            {
                button.setOnClickListener(new View.OnClickListener()
                {
                    @Override
                    public void onClick(View view)
```



```
        {
            handleButtonClick(view.getId());
        }
    }

    bac.setOnClickListener(new View.OnClickListener()
    {
        @Override
        public void onClick(View view)
        {
            t1.setText("");
        }
    });

    private void handleButtonClick(int id)
    {
        String txt = t1.getText().toString();
        if (id == R.id.b16)
        {
            if (!firstNumber.isEmpty() && !operator.isEmpty())
            {
                String secondNumber = txt;
                double result = 0;
                double num1 = Double.parseDouble(firstNumber);
                double num2 = Double.parseDouble(secondNumber);

                switch (operator)
                {
                    case "+":
                        result = num1 + num2;
                        break;
                    case "-":
                        result = num1 - num2;
                        break;
                    case "x":
                        result = num1 * num2;
                        break;
                    case "/":
                        result = num1 / num2;
                        break;
                }
            }
        }
    }
}
```



```

        t1.setText(String.valueOf(result));
        firstNumber = "";
        operator = "";
    }}
    else if (id == R.id.b10)
        if (!txt.contains("."))
t1.setText(txt + ((Button) findViewById(id)).getText());
    else if (id == R.id.b12 || id == R.id.b13 || id == R.id.b14 || id == R.id.b15)
        if (!txt.isEmpty())
        {
            firstNumber = txt;
            operator = ((Button) findViewById(id)).getText().toString();
            t1.setText("");
        }
    else if (id == R.id.b11)
        if (txt.length() > 0)
            t1.setText(txt.substring(0, txt.length() - 1));
    else
        t1.setText(txt + ((Button) findViewById(id)).getText());
    }}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#000000"
    tools:context=".MainActivity">

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:columnCount="4"
        android:rowCount="6"
        android:layout_margin="5dp">

        <EditText
            android:id="@+id/t1"

```

```

            android:layout_marginTop="350dp"
            android:layout_marginBottom="15dp"
            android:layout_width="match_parent"
            android:layout_height="150dp"
            android:layout_row="0"
            android:layout_column="0"
            android:ems="10"
            android:layout_columnSpan="4"
            android:inputType="text"
            android:textSize="92px"
            android:textAlignment="viewEnd"
            android:background="#20ffffff"
            android:textColor="#ffffff"
            android:text=""/>
            android:background="@drawable/button_background"
            android:gravity="center"
            android:text="2"
            android:textAlignment="center"
            android:textSize="22dp"
            android:textStyle="bold"

```



```
android:layout_marginBottom="10dp"/>
```

```
<Button
```

```
    android:id="@+id/b3"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_row="1"
```

```
    android:layout_column="2"
```

```
    android:layout_marginLeft="5dp"
```

```
    android:background="@drawable/button_back  
ground"
```

```
    android:text="3"
```

```
    android:gravity="center"
```

```
    android:textAlignment="center"
```

```
    android:textSize="28dp"
```

```
    android:textStyle="bold"
```

```
    android:layout_marginBottom="10dp"/>
```

```
<Button
```

```
    android:id="@+id/b12"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_marginLeft="5dp"
```

```
    android:layout_row="1"
```

```
    android:layout_column="3"
```

```
    android:gravity="center"
```

```
    android:textAlignment="center"
```

```
    android:textSize="28dp"
```

```
    android:textStyle="bold"
```

```
    android:layout_marginBottom="10dp"
```

```
    android:background="@drawable/op_bg"
```

```
    android:textColor="#ffffff"
```

```
    android:text="+"/>
```

```
<Button
```

```
    android:id="@+id/b4"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_marginLeft="5dp"
```

```
    android:layout_row="2"
```

```
    android:layout_column="0"
```

```
    android:gravity="center"
```

```
    android:textAlignment="center"
```

```
    android:textSize="28dp"
```

```
    android:textStyle="bold"
```

```
    android:layout_marginBottom="10dp"
```

```
    android:background="@drawable/button_back  
ground" android:text="4"/>
```

```
<Button
```

```
    android:id="@+id/b5"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_marginLeft="5dp"
```

```
    android:layout_row="2"
```

```
    android:layout_column="1"
```

```
    android:gravity="center"
```

```
    android:textAlignment="center"
```

```
    android:textSize="28dp"
```

```
    android:textStyle="bold"
```

```
    android:layout_marginBottom="10dp"
```

```
    android:background="@drawable/button_back  
ground" android:text="5"/>
```

```
<Button
```

```
    android:id="@+id/b6"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_marginLeft="5dp"
```

```
    android:layout_row="2"
```

```
    android:layout_column="2"
```

```
    android:background="@drawable/button_back  
ground" android:gravity="center"
```

```
    android:textAlignment="center"
```

```
    android:textSize="28dp"
```

```
    android:textStyle="bold"
```

```
    android:layout_marginBottom="10dp"
```

```
    android:text="6"/>
```

```
<Button
```

```
    android:id="@+id/b13"
```

```
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
```

```
    android:layout_marginLeft="5dp"
```



```

android:layout_row="2"

    android:layout_column="3"
android:background="@drawable/op_bg"
    android:textColor="#ffffff"
android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
    android:text="-"/>

```

<Button

```

    android:id="@+id/b7"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_marginLeft="5dp"
    android:layout_row="3"
    android:layout_column="0"
    android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="7"/>

```

<Button

```

    android:id="@+id/b8"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_marginLeft="5dp"
    android:layout_row="3"
    android:layout_column="1"
    android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="8"/>

```

<Button

```

    android:id="@+id/b9"

```

```

    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_marginLeft="5dp"
    android:layout_row="3"
    android:layout_column="2"
    android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="9"/>

```

<Button

```

    android:id="@+id/b14"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_marginLeft="5dp"
    android:layout_row="3"
    android:layout_column="3"
    android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
android:background="@drawable/op_bg"
    android:textColor="#ffffff"
    android:text="x"/>

```

<Button

```

    android:id="@+id/b0"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_marginLeft="5dp"
    android:layout_row="4"
    android:layout_column="0"
    android:gravity="center"
    android:textAlignment="center"
    android:textSize="28dp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="0"/>

```

<Button


```

android:id="@+id/b10"
android:layout_width="105dp"
android:layout_height="75dp"
android:layout_marginLeft="5dp"
android:layout_row="4"
android:layout_column="1"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp"

```

<Button

```

android:id="@+id/b10"
android:layout_width="105dp"
android:layout_height="75dp"
android:layout_marginLeft="5dp"
android:layout_row="4"
android:layout_column="1"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp"

```

```

android:background="@drawable/button_back
ground" android:text="." />

```

<Button

```

android:id="@+id/b11"
android:layout_width="105dp"
android:layout_height="75dp"
android:layout_marginLeft="5dp"
android:layout_row="4"
android:layout_column="2"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp"

```

```

android:background="@drawable/button_back
ground" android:text="bp" />

```

<Button

```

android:id="@+id/b15"
android:layout_width="105dp"

```

```

android:layout_height="75dp"
android:layout_marginLeft="5dp"
android:layout_row="4"
android:layout_column="3"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp"

```

```

android:background="@drawable/op_bg"
android:textColor="#ffffff"
android:text="/" />

```

<Button

```

android:id="@+id/bac"
android:layout_width="215dp"
android:layout_height="75dp"
android:layout_row="5"
android:layout_column="0"
android:layout_columnSpan="2"
android:layout_marginLeft="5dp"
android:text="AC"
android:textColor="#ffffff"

```

```

android:background="@drawable/op_bg"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp" />

```

<Button

```

android:id="@+id/b16"
android:layout_width="215dp"
android:layout_height="75dp"
android:layout_marginLeft="5dp"
android:layout_row="5"
android:layout_column="2"
android:layout_columnSpan="2"
android:gravity="center"
android:textAlignment="center"
android:textSize="28dp"
android:textStyle="bold"
android:layout_marginBottom="10dp"

```


<code>android:background="@drawable/op_bg"</code>	<code></GridLayout></code>
<code>android:textColor="#ffffff"</code>	<code></LinearLayout></code>
<code>android:text=""/></code>	

Result: The program has executed successfully and output is verified

Date:

APP 3: Intent

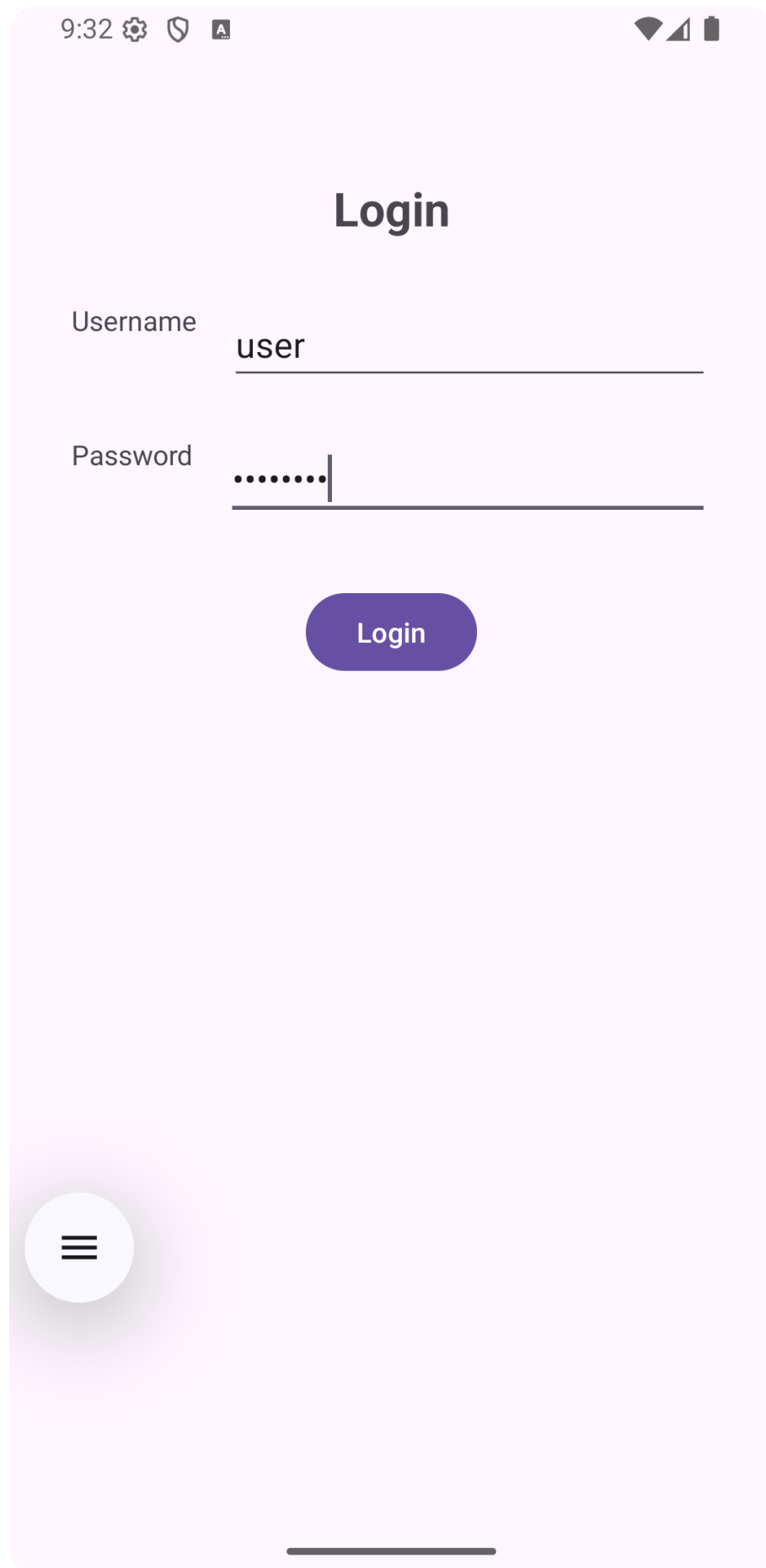
Aim: Create two activities , 1st activity contains login details, when clicked to submit, go to 2nd activity.

Program Code:

MainActivity.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    Button b1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        e1=(EditText) findViewById(R.id.edittext1);
        e2=(EditText) findViewById(R.id.edittext2);
        b1=(Button) findViewById(R.id.btn1);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username=e1.getText().toString().trim();
                String pass=e2.getText().toString();
```



A mobile application login screen mockup. At the top, a status bar shows the time 9:32, a gear icon for settings, a shield icon for security, and a battery icon. Below the status bar, the word "Login" is centered in a large, bold, black font. Underneath, there are two input fields. The first is labeled "Username" and contains the text "user". The second is labeled "Password" and contains seven dots, with a vertical cursor line at the end. Below the password field is a purple rounded rectangular button with the word "Login" in white. At the bottom left, there is a white circular button with a black hamburger menu icon (three horizontal lines). A thin horizontal line is visible at the very bottom of the screen, representing the home indicator bar.

9:32

9:32

Login

Username

user

Password

.....

Login

≡

```

if(username.equals("user") && pass.equals("admin123")) {
    Toast.makeText(MainActivity.this, "Login successful", Toast.LENGTH_SHORT).show();
    Intent intent = new Intent(MainActivity.this, Homepage.class);
    startActivity(intent);
}
else
    Toast.makeText(MainActivity.this,"Login failed",Toast.LENGTH_SHORT).show();
}
});
}
}

```

Activity_Main.xml

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="64dp" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Username"
        app:layout_constraintTop_toBottomOf="@id/textView3"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="32dp"
```

```

        android:layout_marginStart="32dp" />
<EditText
    android:id="@+id/edittext1"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter username"
    android:inputType="text"
app:layout_constraintStart_toEndOf="@id/textView4"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textView4"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="32dp" />
<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Password"
app:layout_constraintTop_toBottomOf="@id/edittext1"
app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="24dp"
    android:layout_marginStart="32dp" />
<EditText
    android:id="@+id/edittext2"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter password"
    android:inputType="textPassword"

```

9:33   



Welcome to the Homepage!

Logout



Login successful

```

app:layout_constraintStart_toEndOf="@id/textView5"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textView5"

    android:layout_marginStart="16dp"
    android:layout_marginEnd="32dp" />

<Button
    android:id="@+id/btn1"
    android:layout_width="wrap_content"

```

```

    android:layout_height="wrap_content"
    android:text="Login"
app:layout_constraintTop_toBottomOf="@id/editText2"
app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Homepage.java

```

package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Homepage extends AppCompatActivity {
    Button logoutButton;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_homepage);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        logoutButton = findViewById(R.id.logoutButton);
        logoutButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Go back to login screen
            }
        });
    }
}

```



```

        Intent intent = new Intent(Hompage.this, MainActivity.class);
        startActivity(intent);
        finish();
    }
});
}}
```

Activity_HomePage.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/main"
    android:padding="24dp">

    <TextView
        android:id="@+id/welcomeText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to the
Homepage!"
        android:textSize="20sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="64dp" />

    <Button
        android:id="@+id/logoutButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Logout"
```

```

        app:layout_constraintTop_toBottomOf="@id/
welcomeText"

        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="32dp" />
    </androidx.constraintlayout.widget.Constraint
Layout>
```

Result: The program has executed successfully and output is verified

Date :

APP 4: Activity Life Cycle

Aim: Create an app to demonstrate Activity Life Cycle.

Program Code :

MainActivity.java

```
package com.example.start;
import androidx.appcompat.app.AppCompatActivity;
import android.nfc.Tag;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends
AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d("Appstate","Created");
    }
    @Override
    protected void onStart()
    {
        super.onStart();
        Log.d("Appstate","Starting");
    }
    @Override
    protected void onResume()
    {
        super.onResume();
        Log.d("Appstate","Resuming");
    }
```

```
@Override
protected void onPause()
{
    super.onPause();
    Log.d("Appstate","Paused");
}
@Override
protected void onStop()
{
    super.onStop();
    Log.d("Appstate","Stopped");
}
@Override
protected void onRestart()
{
    super.onRestart();
    Log.d("Appstate","Restarted");
}
@Override
protected void onDestroy()
{
    super.onDestroy();
    Log.d("Appstate","Destroyed");
}
```

Result: The program has executed successfully and output is verified

Date :

APP 5: Checkbox and Radiobutton

Aim : Create an app to demonstrate the usage of Checkbox and Radio buttons

Program Code :

MainActivity.java:

```
package com.example.radiocheck;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity
{
    Button b1;
    RadioGroup rg;
    RadioButton r1,r2;
    CheckBox e,m,h;
    TextView t;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        b1=(Button) findViewById(R.id.submit);
        r1=(RadioButton) findViewById(R.id.male);
        r2=(RadioButton) findViewById(R.id.female);
        e=(CheckBox)findViewById(R.id.e);
        m=(CheckBox)findViewById(R.id.m);
        h=(CheckBox)findViewById(R.id.h);
        t=(TextView)findViewById(R.id.tv);
        b1.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View view)
            {
                String s[]=new String[3];
                t.setText("Languages Known: ");
                if(e.isChecked())
                t.setText(t.getText().toString()+" "+e.getText());
```

```
if(m.isChecked())  
t.setText(t.getText().toString()+" "+m.getText());
```

```

if(h.isChecked())
t.setText(t.getText().toString()+" "+h.getText());
});
rg=(RadioGroup) findViewById(R.id.rg);
rg.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener()
{
@Override
public void onCheckedChanged(RadioGroup group, int checkedId)
{
if(checkedId==R.id.female)
{ Toast.makeText(getApplicationContext(),r2.getText().toString(),Toast.LENGTH_SHORT).show();
}
else
{ Toast.makeText(getApplicationContext(),r1.getText().toString(),Toast.LENGTH_SHORT).show();
}
}
});
}
}
}

```

Activity_home_page.xml

```

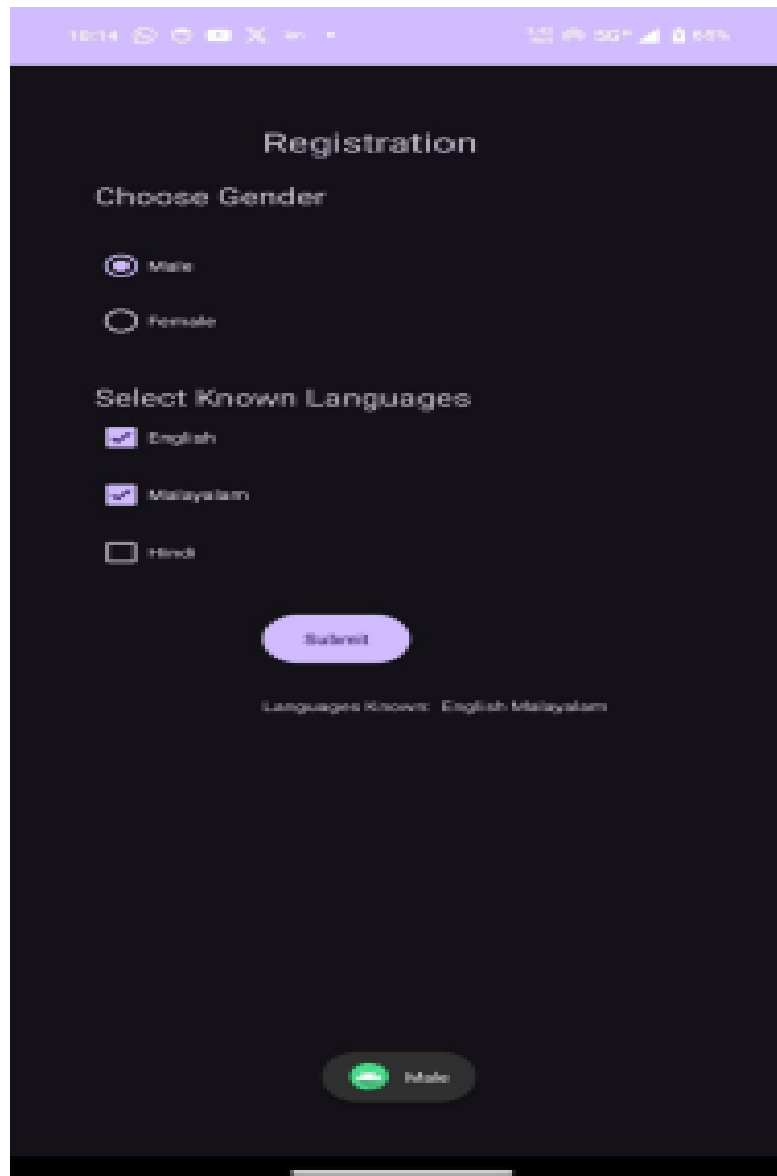
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="match_parent"
android:layout_height="match_parent">
<TextView
android:layout_width="wrap_content"
android:layout_height="30dp"
android:layout_marginTop="50dp"
android:layout_marginLeft="150dp"
android:id="@+id/h1"
android:text="Registration"
android:textSize="24dp"/>
<TextView
android:layout_width="wrap_content"

```

```

android:layout_height="30dp"
android:layout_marginTop="100dp"
android:layout_marginLeft="50dp"
android:id="@+id/gender"
android:text="Choose Gender"
android:textSize="20dp"/>
<RadioGroup
android:layout_width="250dp"
android:layout_height="wrap_content"
android:id="@+id/rg"
android:layout_marginTop="150dp"
android:layout_marginLeft="50dp">
<RadioButton
android:layout_width=
"wrap_content"
android:layout_height=
"wrap_content"
android:text="Male"
android:id="@+id/male"/>
<RadioButton
android:layout_width=
"wrap_content"
android:layout_height=
"wrap_content"

```



A screenshot of a mobile application registration screen. The screen has a dark blue background. At the top, there is a status bar with the time 10:14, signal strength, 5G+ network, and 55% battery. The title "Registration" is centered at the top in white. Below it, the text "Choose Gender" is displayed. There are two radio button options: "Male" (selected) and "Female". Below this, the text "Select Known Languages" is displayed. There are three checkbox options: "English" (checked), "Malayalam" (checked), and "Hindi" (unchecked). A blue "Submit" button is centered below the language options. Below the button, the text "Languages Known: English Malayalam" is displayed. At the bottom of the screen, there is a toggle switch for "Male" which is currently turned on.

Registration

Choose Gender

☒ Male

☐ Female

Select Known Languages

☒ English

☒ Malayalam

☐ Hindi

Submit

Languages Known: English Malayalam

☒ Male

```
android:text="Female"
android:id="@+id/female"/>
</RadioGroup>
<TextView
android:layout_width="wrap_content"
android:layout_height="30dp"
android:layout_marginTop="275dp"
android:layout_marginLeft="50dp"
android:id="@+id/lang"
android:text="Select Known Languages"
android:textSize="20dp"/>
<CheckBox
android:id="@+id/e"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="English"
android:layout_marginTop="300dp"
android:layout_marginLeft="50dp"/>
<CheckBox
android:id="@+id/m"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Malayalam"
android:layout_marginTop="350dp"
android:layout_marginLeft="50dp"/>
<CheckBox
android:id="@+id/h"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hindi"
android:layout_marginTop="400dp"
android:layout_marginLeft="50dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="50dp"
android:layout_marginTop="475dp"
android:layout_marginLeft="150dp"
android:text="Submit"
android:id="@+id/submit"/>
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="50dp"
android:layout_marginTop="550dp"
android:layout_marginLeft="150dp"
android:text=""
android:id="@+id/tv"/>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.

Date :

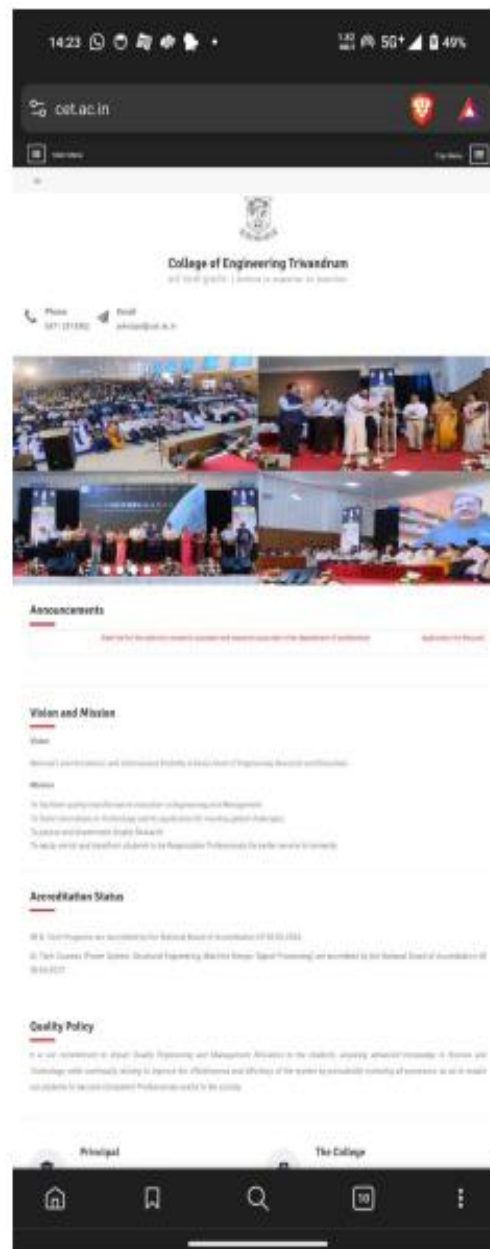
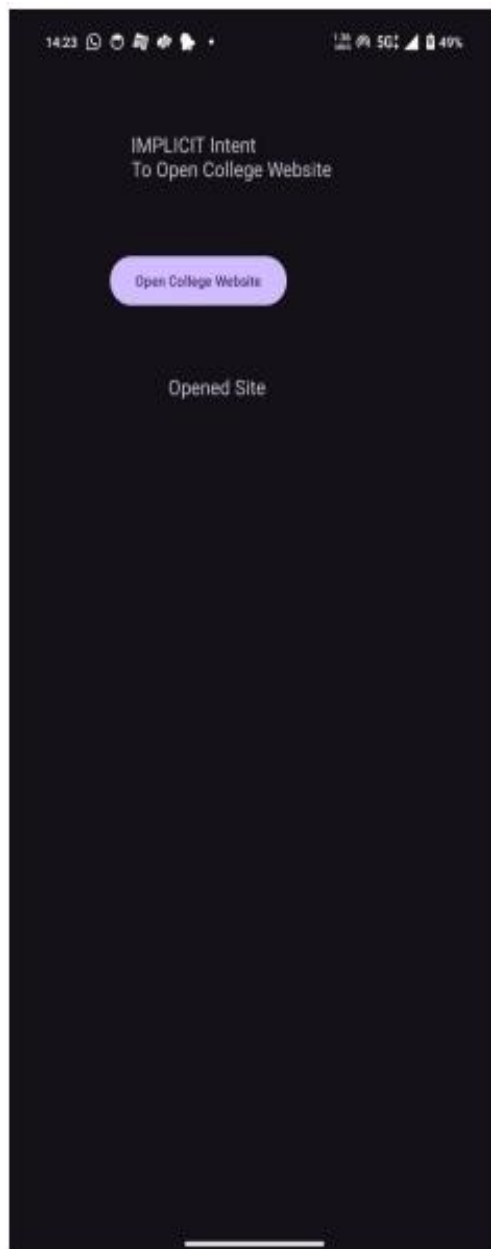
APP 6: Implicit Intent

Aim : Create an app to demonstrate the usage of implicit intent by opening a website from the app

Program Code :

MainActivity.java:

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
    Button b;
    TextView t;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        b=(Button) findViewById(R.id.b1);
        t=(TextView) findViewById(R.id.t1);
        b.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("https://www.cet.ac.in"));
                startActivity(intent);
                t.setText("Youtube Opened");
            }
        });
    }
}
```



Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <RelativeLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="IMPLICIT Intent \nTo
Open College Website"
            android:layout_marginLeft="115dp"
            android:layout_marginTop="100dp"
            android:textSize="20sp">
        </TextView>

        <Button
            android:id="@+id/b1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginLeft="95dp"
            android:layout_marginTop="200dp"
            android:text="Open College Website">
        </Button>
```

```
<TextView
    android:id="@+id/t1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text=""
    android:layout_marginTop="300dp"
    android:layout_marginLeft="150dp"
    android:textSize="20sp"></TextView>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

Date :

APP 7: List View

Aim : Create an app to create a List view with few items in it

Program Code :

MainActivity.java:

```
package com.example.listview;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.OnBackPressedCallback;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
    ListView lv;
    String arr[]={"Deepak","Subi","Misira","Bhavana","Zuhri"};
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        lv=(ListView) findViewById(R.id.lv);
        ArrayAdapter adapter=new ArrayAdapter<String>(this,
        android.R.layout.simple_list_item_1,arr);
        lv.setAdapter(adapter);
        lv.setOnItemClickListener(new AdapterView.OnItemClickListener()
        {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id)
            {
                Toast.makeText(MainActivity.this,"Name Clicked: "+adapter.getItem(position),
                Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

}

5:02   

LIST VIEW

Deepak

Subi

Misira

Bhavana

Zuhri

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:layout_marginLeft="100dp"
android:text="LIST VIEW"
android:textSize="42dp">
</TextView>

<ListView
android:id="@+id/lv"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="115dp"
android:layout_marginBottom="50dp"
android:layout_marginLeft="10dp">
</ListView>

</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.

Date :

APP 8 : Scroll View

Aim : Create an app to create a spinner and select an item from it.

Program Code :

MainActivity.java:

```
package com.example.scrollview;
import android.os.Bundle;
import android.widget.LinearLayout;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        LinearLayout linearLayout = findViewById(R.id.linearLayoutContainer);
        for (int i = 1; i <= 20; i++)
        {
            TextView textView = new TextView(this);
            textView.setText("TextView " + i);
            textView.setTextSize(45);
            textView.setPadding(16, 16, 16, 16);
            linearLayout.addView(textView);
        }
    }
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">
    <LinearLayout
        android:id="@+id/linearLayoutContainer"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">
    </LinearLayout>
</ScrollView>
```

Result: The program has executed successfully and output is verified



Date :

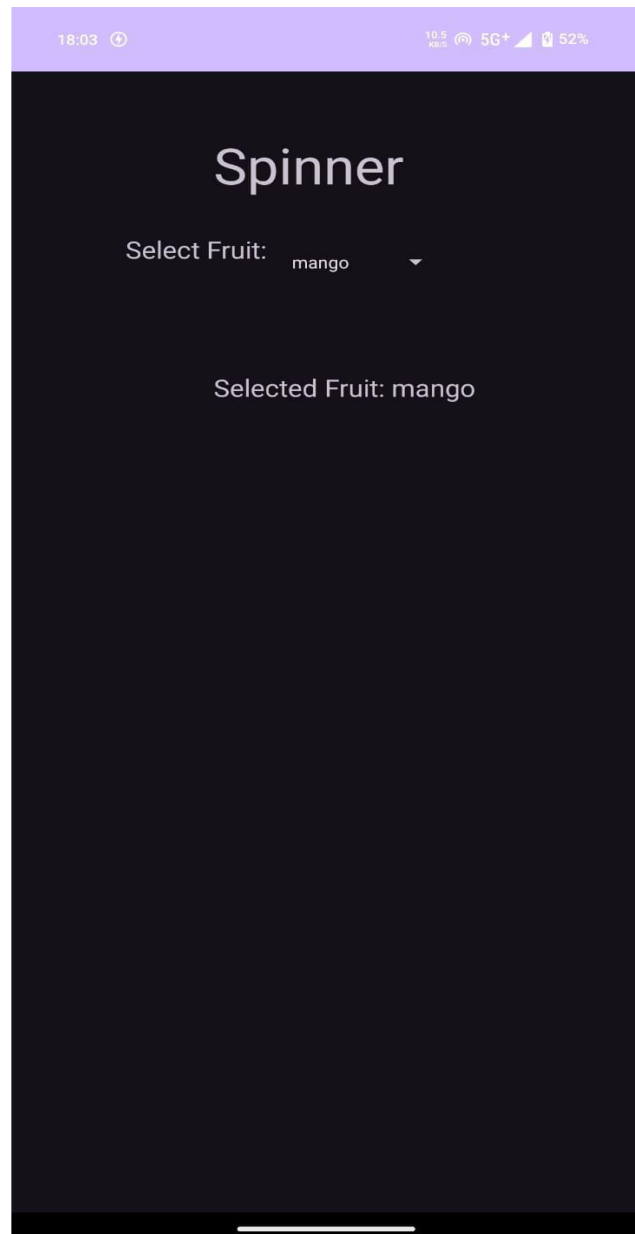
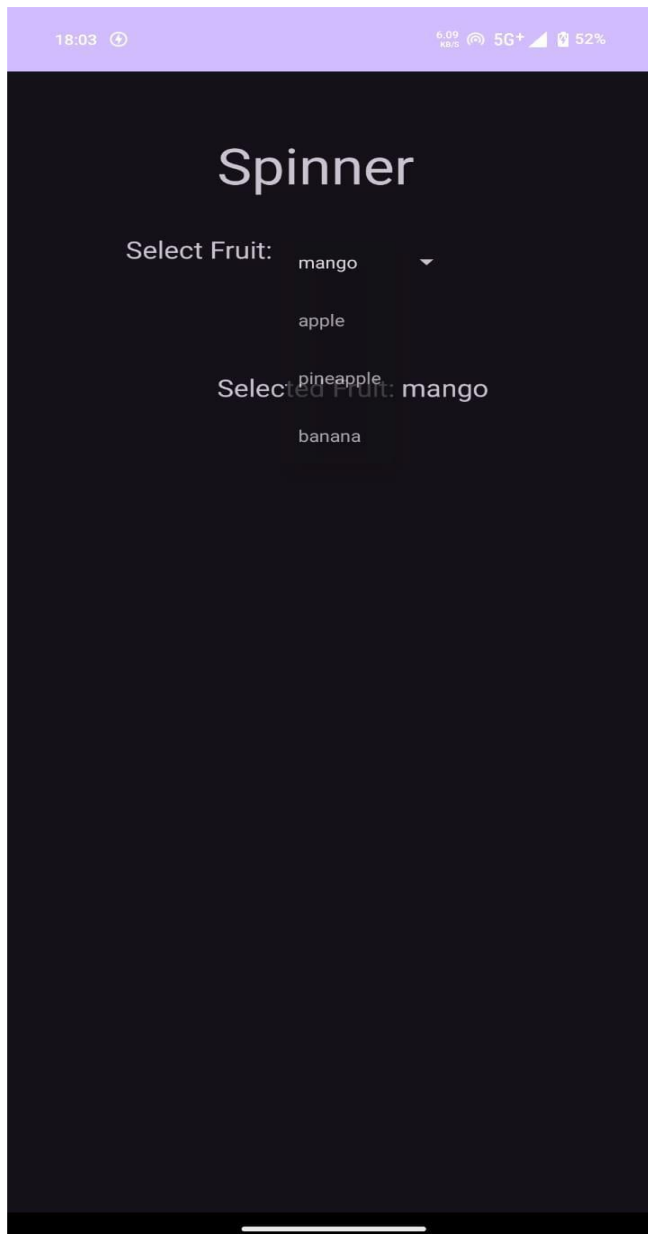
APP 9 : Spinner

Aim : Create an app to create a spinner and select an item from it.

Program Code :

MainActivity.java:

```
package com.example.spinner;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity
{
    Spinner s;
    TextView t1;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        s=(Spinner)findViewById(R.id.sp);
        t1=(TextView) findViewById(R.id.t1);
        ArrayList <String> al=new ArrayList<String>();
        al.add("mango");
        al.add("apple");
        al.add("pineapple");
        al.add("banana");
        ArrayAdapter ad=new ArrayAdapter<>(this, androidx.appcompat.R.layout.support_simple_spinner_dropdown_item,al);
        s.setAdapter(ad);
        s.setOnItemClickListener(new AdapterView.OnItemClickListener()
        {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id)
            {
                t1.setText("Selected Fruit: "+al.get(position));
            }
        })
    }
}
```



```

@Override
public void onNothingSelected(AdapterView<?> parent)
{
    t1.setText("Select a Fruit !!!!");
}
});
}
}
}

```

Activity_home_page.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:text="Spinner"
android:textSize="42dp"></TextView>
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="85dp"
android:layout_marginTop="135dp"
android:text="Select Fruit:"
android:textSize="20dp">
</TextView>

```

```

<Spinner
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="200dp"
android:layout_marginTop="135dp"
android:id="@+id/sp"></Spinner>
<TextView
android:id="@+id/t1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="250dp"
android:text=""
android:textSize="20dp"></TextView>
</RelativeLayout>
</LinearLayout>

```

Result: The program has executed successfully and output is verified.

Date :

APP 10 : Shared Preferences

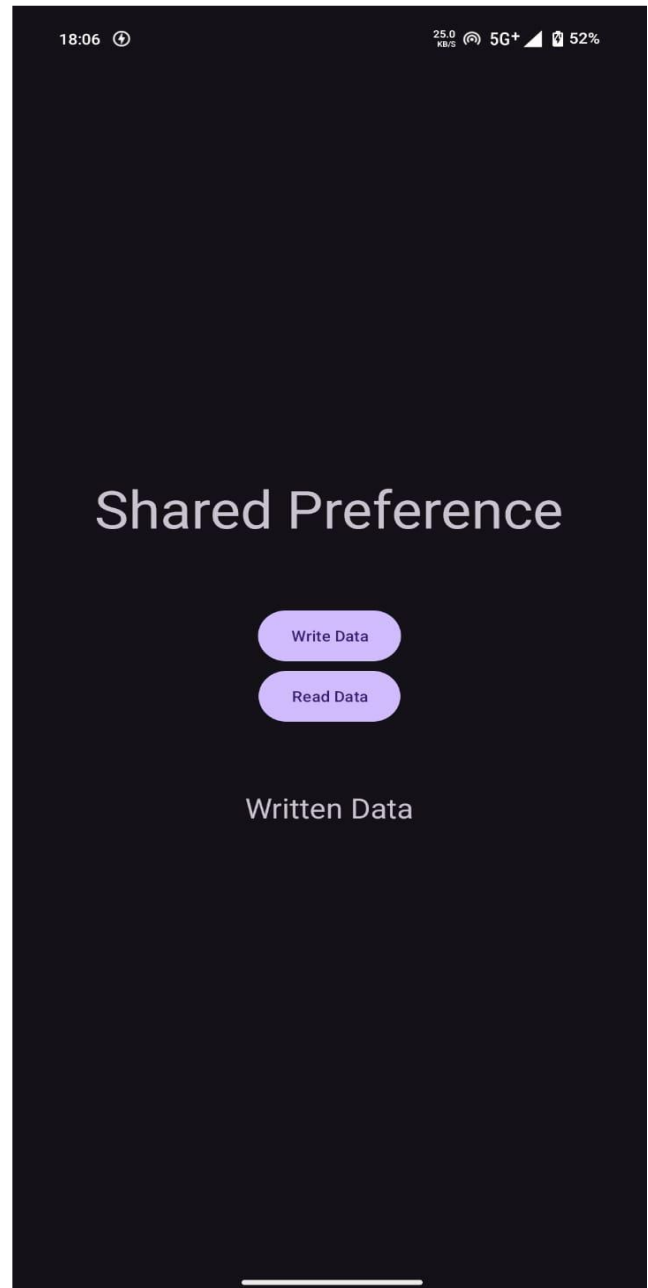
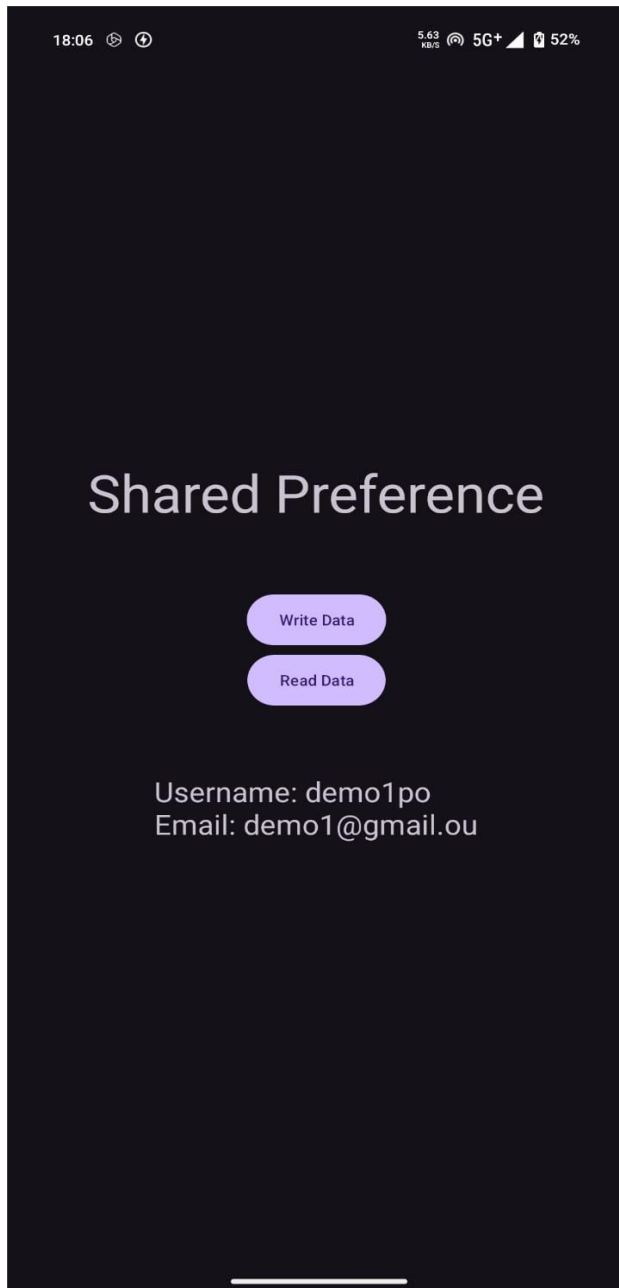
Aim : Create an app to demonstrate the usage of shared preferences by writing and reading data.

Program Code :

MainActivity.java

```
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
    TextView txt;
    Button br,bw;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        txt=(TextView) findViewById(R.id.text_status);
        bw=(Button) findViewById(R.id.btn_write);
        br=(Button) findViewById(R.id.btn_read);
        String usernamedemo="demo1po";
        String emaildemo="demo1@gmail.ou";
        SharedPreferences sp =getSharedPreferences("sp_abc",MODE_PRIVATE);
        bw.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View view)
            {
                SharedPreferences.Editor se = sp.edit();
                se.putString("username",usernamedemo);
                se.putString("email",emaildemo);
                se.apply();
                txt.setText("Written Data");
            }
        });
    }
}
```

```
}  
});
```



```
br.setOnClickListener(new View.OnClickListener()
{
    @Override
    public void onClick(View view) {
        String u,e;
        u=sp.getString("username","");
        e=sp.getString("email","");
        txt.setText("Username: "+u+" \nEmail: "+e);
    }
});
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"
android:orientation="vertical"
android:gravity="center">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Shared Preference"
android:textSize="42dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Write Data"
android:layout_marginTop="50dp"
android:id="@+id/btn_write"/>
```

```
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Read Data"
android:id="@+id/btn_read"/>
<TextView
android:id="@+id/text_status"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="STATUS"
android:textSize="22dp"/>

</LinearLayout>
```

Result: The program has executed successfully and output is verified.

